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City-Scale Industrial Land Grounding and Environmental Risk Mapping with Retrieval-Augmented Vision-Language Models

Yuling Wu, Xiyu Jin, Ruiqian Zhang*, and Qingyu Li*

***Abstract*—Effective management of industrial land is essential given its substantial environmental footprint, intensive resource consumption, and implications for urban planning and public health. Accurate visual grounding of industrial parcels in remote sensing imagery provides a critical basis for assessing associated environmental risks and supporting evidence-based resource allocation. However, existing approaches often rely primarily on visual cues and remain vulnerable to domain-specific ambiguity in complex urban environments. To address these limitations, we propose Retrieval-Augmented Geospatial Vision-Language Grounding (RAGVLG), a model-agnostic framework that integrates structured knowledge retrieval with vision-language models (VLMs) to improve industrial land grounding in remote sensing imagery. RAGVLG retrieves domain knowledge and contextual exemplars to guide grounding decisions, enabling flexible adaptation to different VLM backbones. We construct an industrial land-grounding dataset and systematically evaluate the proposed framework, demonstrating consistent improvements over baseline VLMs across multiple backbones. We further demonstrate its utility through city-scale applications in Shenzhen and Hong Kong, where industrial land is mapped, and the spatial distribution of potential environmental risks is assessed by integrating key environmental parameters. Local Indicators of Spatial Association (LISA) are used to quantify spatial clustering of risk. The results show that High–High hotspots account for 10.62% of all grids in Shenzhen, forming a continuous coastal corridor primarily across Nanshan and Baoan Districts. Hong Kong exhibits a higher proportion of High–High clusters, at 15.31%, with more fragmented high-risk areas concentrated in Kwai Tsing and Kowloon. These findings provide quantitative and spatially explicit evidence for regional industrial planning, environmental risk management, and sustainable industrial development. The dataset and code will be made publicly available at https://github.com/Doris5102023/RAGVLG-2025.**

***Index Terms*—Remote sensing, Retrieval-augmented generation, Vision-language models, Visual grounding, Industrial land**

I. Introduction

Industrial activities such as manufacturing, processing and logistics play a vital role in boosting global economic growth, yet they generate considerable environmental and social pressures, including the urban heat island (UHI) effect, pollution, and rising public health risks. Addressing the sustainability challenges of industrial expansion aligns closely with the United Nations Sustainable Development Goals (SDGs), which call for strategies that reconcile economic development with social well-being and ecological integrity [1]). For instance, SDG 3 targets health issues such as heat-related illnesses and workplace accidents prevalent in industrial zones [2]; SDG 7 emphasizes energy efficiency, especially in "high-heat, low-green" industrial areas prioritized for green transformation [3]; SDG 9 encourages the growth of resource-efficient and resilient industrial facilities [4]; and SDG 13 underscores the importance of heat mitigation and greening initiatives to improve resilience in vulnerable industrial districts [5].

The environmental footprint of industrial activities is closely tied to the extent, spatial distribution, and management of land designated or developed for industrial use. Achieving sustainability goals, therefore, requires accurate identification and monitoring of industrial land and its spatial patterns. Remote sensing imagery offers a valuable data source in this context: high-resolution images capture detailed spatial, geometric, and textural features, enabling precise visual delineation of industrial land [6].

A variety of methods have been developed to extract land-use information from remote sensing imagery. Traditional techniques include (i) manual visual interpretation based on textural and structural features [7], (ii) statistical classification using spectral features [8], and (iii) thematic extraction via index models such as the normalized difference built-up index (NDBI) [9]. However, in complex urban environments these methods are often constrained by spectral confusion, mixed pixels, and heterogeneous land-use patterns, highlighting the need for more robust, scalable, and intelligent monitoring approaches. With the rapid development of artificial intelligence, deep learning has become the dominant paradigm, enabling end-to-end inference with improved accuracy and efficiency. Existing methods can be broadly categorized by input modality. Some methods take remote sensing imagery as the sole input and refine network designs to make full use of inherent visual features [10], [11], [12]. Others combine auxiliary data such as GIS vectors [13], [14], points of interest [15], [16], [17], and social sensing data [18], [19] to characterize spatial structure, functional semantics, and dynamic land use patterns. These multimodal strategies focus on fusing heterogeneous datasets and perform well in complex, heterogeneous settings.

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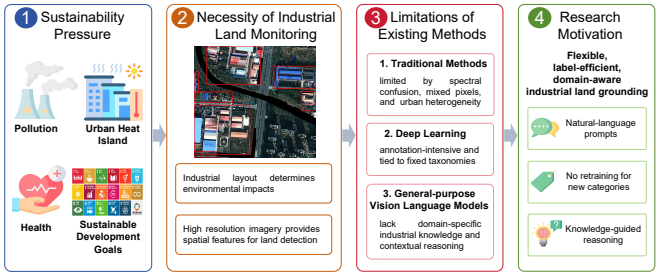


Fig. 1. Motivation diagram

Nevertheless, most deep learning approaches still depend on large volumes of high-quality annotated data, which remain costly and time-consuming to obtain at scale.

To reduce the reliance on manually labeled data, recent studies have increasingly turned to label-efficient and zero-shot learning paradigms. Vision-language models (VLMs), pretrained on large-scale image-text corpora, offer a promising solution by enabling open-vocabulary recognition and substantially reducing annotation requirements. However, when applied to industrial land recognition in remote sensing imagery, their performance is often constrained by the absence of domain-specific knowledge in general-purpose pretraining data, including infrastructure types, production-related layout patterns, zoning regulations, and spatial dependencies. This knowledge gap makes it difficult for VLMs to disambiguate visually similar yet functionally distinct urban scenes, such as industrial facilities, storage areas, logistics sites, and dense urban villages, thereby limiting their practical reliability in complex urban environments [20]. This limitation is further amplified by the context-dependent nature of land use classification. The semantic granularity required for industrial land analysis varies substantially across applications [21]. Regional monitoring may only require a broad industrial/non-industrial distinction, whereas urban planning, environmental regulation, or industrial upgrading may demand more detailed categories, such as heavy manufacturing, electronics production, logistics and warehousing, and creative industry zones. These heterogeneous and evolving analytical needs are difficult to satisfy with a single fixed classification scheme. Yet, most existing approaches remain tied to predefined taxonomies and supervised learning pipelines that depend on large labeled datasets, thereby restricting models to a closed set of categories. As a result, adapting them to new regions, tasks, or semantic definitions often requires costly relabeling and retraining, limiting their transferability, scalability, and operational applicability. These challenges highlight the need for a more flexible knowledge-guided framework capable of incorporating domain knowledge and adapting to diverse industrial land interpretation tasks. The research motivation diagram for this paper is shown in Figure 1.

To overcome these obstacles, we propose a language-driven visual–semantic alignment framework that uses natural language as a flexible semantic interface. Users specify target land-use types via text prompts, which are encoded into semantic representations and aligned with visual features extracted from remote sensing imagery. This allows dynamic grounding of user-defined categories without retraining, overcoming the limitations of fixed taxonomies and reducing dependence on manual annotation, while improving adaptability across diverse application contexts and mitigating domain gaps in general-purpose VLMs. To strengthen contextual and semantic reasoning, we integrate a structured knowledge retrieval module that augments visual understanding with domain-aware information. Specifically, we adopt a graph-based retrieval approach [22] to organize knowledge including subdivision types, ownership attributes, spatial thresholds, and boundary rules into entity–attribute–relation graphs. At inference time, the VLM provides visual features while the retrieval module supplies relevant structured knowledge, enabling spatial–semantic alignment and knowledge-informed reasoning. For rigorous evaluation, we construct the RSIL dataset, containing 1,389 remote sensing images with annotated industrial land bounding boxes, offering a dedicated benchmark for visual grounding of industrial land.

Building on this framework, we select Shenzhen and Hong Kong as the study areas to examine the spatial dynamics of industrial land in relation to urban thermal and ecological conditions. After extracting industrial parcels from high-resolution imagery, we integrate land surface temperature (LST) and vegetation coverage data to perform a comprehensive geospatial analysis. We focus on identifying industrial parcels that are both hotter and less vegetated than their surroundings and designate these as priority targets for greening, cooling, and permeable surface retrofits. In doing so, the study not only advances methods for industrial land monitoring and management but also provides actionable guidance to enhance urban resilience, ecological quality, and community well-being.

Our key contributions are as follows:

1) We propose RAGVLG, a retrieval-augmented vision–language framework for visual grounding of industrial land in remote sensing imagery, enhancing localization accuracy through knowledge-informed inference.

2) We construct a dedicated dataset comprising 1,389 high-resolution images with corresponding ground-truth bounding box annotations of industrial land, supporting rigorous and reproducible evaluation.

3) We demonstrate that RAGVLG is model-agnostic and can be seamlessly adapted to various mainstream VLM backbones, consistently improving industrial land grounding performance across different models.

4) We conduct city-scale case studies in Shenzhen and Hong Kong, performing large-scale industrial land identification and analyzing the spatial distribution of potential environmental risks, thereby providing quantitative evidence to support regional industrial planning and environmental risk management.

II. Data Sets and Study Area

In this study, we construct a regionally heterogeneous dataset named RSIL (Remote Sensing Industrial Land) to assess the efficacy of the proposed RAGVLG framework for

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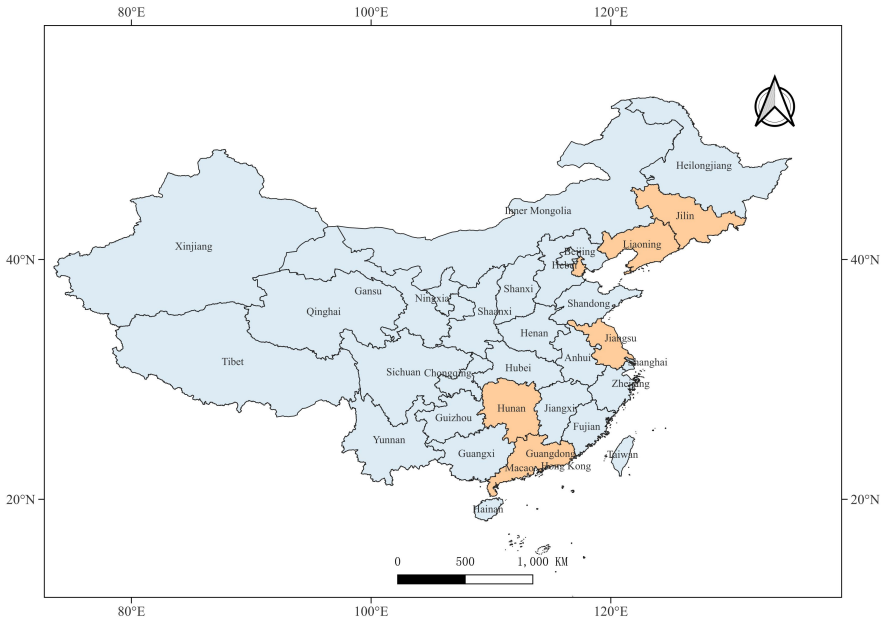


Fig. 2. Geographic location of the study areas in China. Highlighted areas in orange denote the study regions.

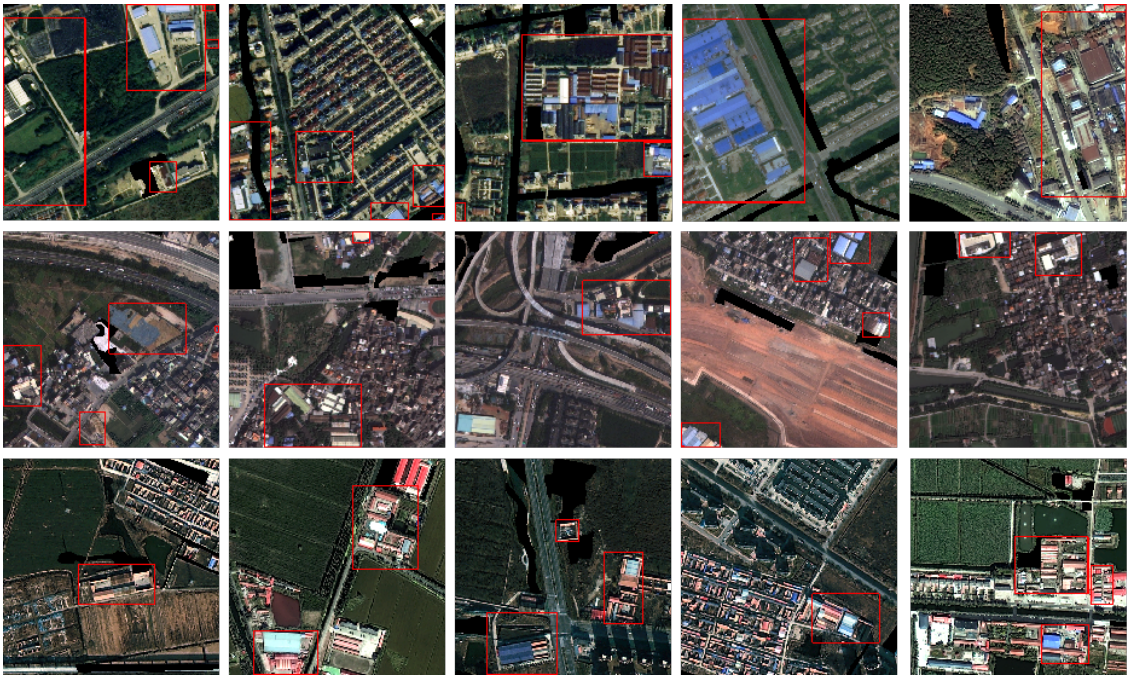


Fig. 3. Remote sensing imagery with ground reference bounding box annotation (in red) of industrial land

visual grounding of industrial land. RSIL is derived from the EVLab-CMCD dataset [23], a large-scale change detection resource that contains pre-change land use maps, post-change remote sensing imagery, and change masks for multiple regions in China.

The remote sensing imagery in RSIL dataset is obtained from the post-change imagery subset of EVLab-CMCD. Afterward, we developed a quick and economical annotation pipeline to extract industrial land parcels from this imagery:

- **Overlay analysis** - pre-change land use maps were overlaid with change masks to isolate areas that had not changed between the two survey periods.
- **Class-based selection** - among these unchanged areas, we selected polygons labeled as industrial land according to the land use maps.
- **Mapping to imagery** - the selected stable industrial parcels were directly mapped onto the corresponding post-change satellite images, forming precise bounding box annotations.

This automated process ensures high annotation reliability by leveraging authoritative land use data, while simultaneously reducing the need for manual labeling.

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The RSIL dataset spans 10 distinct regions across multiple provinces in China, including Jilin, Liaoning, Tianjin, as shown in Figure 2. It comprises 1,389 remote sensing imagery, each with a fixed size of 512×512 pixels and a spatial resolution of 0.8 meters, mostly captured by Chinese satellites such as GaoFen-2 (GF-2) and BeiJing-2 (BJ-2) [23]. Each image is annotated with ground-truth bounding boxes of industrial land parcels. Figure 3 illustrates sample images and annotations from the RSIL dataset.

To further illustrate the practical utility of the RAGVLG framework, we conduct city-scale case studies in two representative megacities in southern China: Shenzhen and Hong Kong. Shenzhen is a core economic hub in Guangdong Province that administers 10 districts with a permanent population of approximately 13.44 million. It relies on industry as its economic pillar and has ranked first nationwide in indicators such as the total output value of industrial enterprises above designated size for three consecutive years since 2022 [24], [25]. Hong Kong is a major international financial and trade center with a unique industrial structure dominated by port-related, logistics, and light industrial activities concentrated in specific districts, providing a valuable comparative case to evaluate the framework's adaptability across diverse urban industrial contexts [26].

The remote sensing imagery adopted in this study, which covers Shenzhen and Hong Kong, is derived from PlanetScope satellites [27], with red, green, and blue (RGB) bands at a spatial resolution of 3 m/pixel. To facilitate scientific management of industrial land, we combine vegetation coverage and thermal environment as core parameters. According to ecological vulnerability, we determine priority intervention areas and focus on low-vegetation and high-heat regions for greening and cooling measures [28]. We calculate NDVI (for vegetation coverage) from Landsat 8 OLI/TIRS imagery [29] using June–August 2023 composite data, and derive LST (for thermal environment) from the same dataset [30]. Combining NDVI and LST enables targeted identification of "hot and low-greenness" industrial zones needing urgent ecological intervention.

III. Methodology

A. Overview

This study develops a two-stage workflow for industrial land identification and environmental risk assessment, as illustrated in Figure 4. The workflow consists of: 1) visual grounding of industrial land using the proposed retrieval-augmented geospatial vision-language grounding framework, and 2) geographic-spatial evaluation of potential environmental risks associated with the identified industrial land.

In the first stage, we evaluate the effectiveness of the proposed RAGVLG framework using the RSIL dataset. RAGVLG integrates VLMs with retrieval-augmented generation to incorporate domain-specific knowledge into remote sensing image interpretation. Specifically, structured documents, unstructured texts, user queries, and remote sensing imagery are jointly processed through a knowledge retrieval and semantic embedding layer, which retrieves relevant domain knowledge and contextual exemplars. These retrieved materials are then incorporated into prompts to guide VLM reasoning, cross-modal semantic fusion, and spatial grounding. This design enables accurate industrial land grounding while reducing dependence on large-scale manually labeled datasets.

In the second stage, the RAGVLG framework is applied to large-scale industrial land identification in Shenzhen and Hong Kong. Based on the resulting industrial land maps, we develop a geospatial risk assessment model to evaluate potential environmental risks. The model uses land surface temperature and vegetation coverage, represented by thermal anomalies and greening deficits, as two key structural indicators. These indicators are integrated to classify risk levels and identify spatial clusters of high-risk industrial areas. The resulting risk maps support the delineation of priority intervention zones and provide spatial evidence for differentiated ecological management, sustainable industrial transformation, and industrial layout optimization.

B. Visual grounding of industrial land

Industrial land refers to spatial areas officially designated or developed for manufacturing, processing, logistics, warehousing, and other industrial activities. In remote sensing images, these areas can be distinguished by characteristic visual features such as large building footprints, hard surfaces, limited vegetation coverage, and organized spatial layouts (roads, loading zones, storage yards). The identification process involves localizing industrial parcels within high resolution imagery using visual grounding techniques.

Our RAGVLG framework addresses two major challenges: 1) Data dependency: Traditional approaches rely on substantial volumes of annotated training data, which are expensive and time-consuming to generate. 2) Domain knowledge gaps: General-purpose models often lack the specialized knowledge necessary for accurately distinguishing industrial land from visually similar categories. To address this, we can extract the formal definition of industrial land from the tabular data of the Third National Land Survey and integrate this authoritative domain knowledge into the model, thereby improving the accuracy of industrial land identification amid visually confounding land use types.

Recent advances in VLMs have introduced powerful zero-shot learning capabilities through large-scale pretraining on massive image–text corpora. VLMs represent a class of multimodal artificial intelligence systems that combine natural language processing and computer vision to jointly model textual and visual information and can learn to align visual features with corresponding semantic representations. In recent years, improvements in architecture design and expansion of pretraining datasets have enabled VLMs to attain state-of-the-art performance across diverse computer vision tasks [31], including visual question answering [32], image understanding [33], semantic segmentation [34], text-to-image generation [35], and object detection [36]. This versatility demonstrates strong generalizability and robustness in real-world scenarios. However, while general-purpose VLMs excel in broad perception tasks, their pretrained knowledge often

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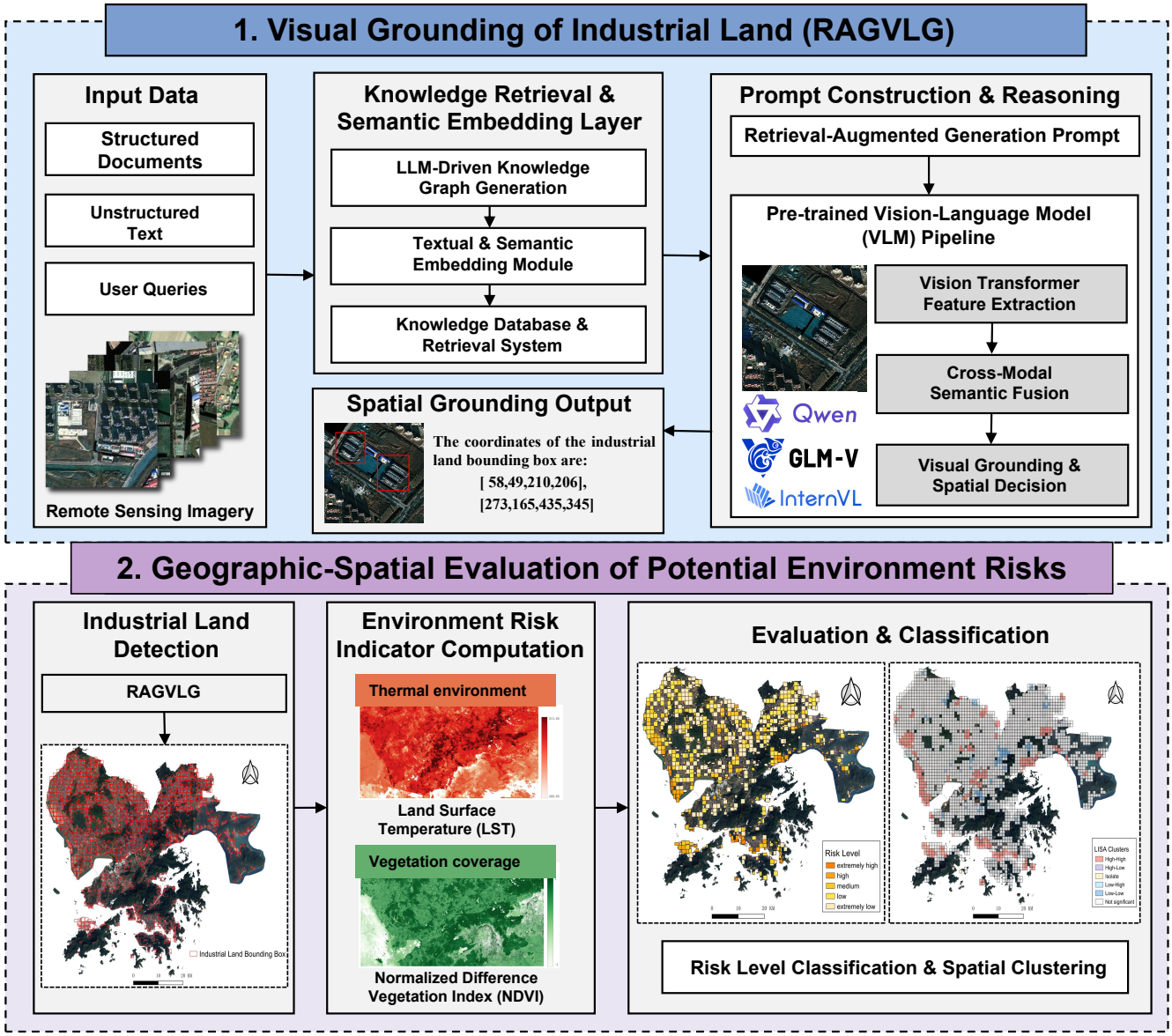


Fig. 4. **Architectural overview of the proposed framework.** The operational pipeline is divided into two primary phases: **Phase 1: Visual Grounding of Industrial Land (RAGVLG)**. This phase inputs texts and remote sensing imagery, performs semantic embedding and LLM-driven knowledge retrieval to construct context-aware prompts, and leverages pretrained VLMs to output initial bounding boxes for industrial parcels. **Phase 2: Geographic-Spatial Evaluation of Potential Environmental Risks**. This phase routes the predicted bounding boxes into a geospatial engine to convert them into actual geographic coordinates, extracts localized LST and NDVI data to calculate comprehensive risk priority scores.

underrepresents fine-grained, domain-specific concepts such as industrial facility types, zoning regulations, and spatial adjacency constraints, which are critical for accurate industrial land recognition.

To address this knowledge gap, we incorporate retrieval-augmented generation (RAG) to supply external domain-specific information. In standard RAG, a retriever searches relevant documents or knowledge items from external databases, which are subsequently combined with the first query to enhance the context. This augmented input is processed by a generator (e.g., a VLM or large language model (LLM)) to generate results that are more precise and semantically coherent [37], [38]. RAG offers strong adaptability, requiring minimal retraining for different tasks [39], and has shown notable potential across natural language processing (NLP) applications [40], [41]. However, vector-based retrieval approaches in traditional RAG frameworks often struggle in scenarios requiring multi-hop reasoning or deep relational inference [42].

To overcome these limitations, we adopt Graph RAG [22], which structures retrieved knowledge as an entity–attribute–relationship graph. By leveraging linked entity networks rather than isolated vectors, Graph RAG strengthens semantic association, supports multi-hop retrieval, and enhances interpretability during inference. In our framework (see Figure 5), the VLM first generates descriptive represen-

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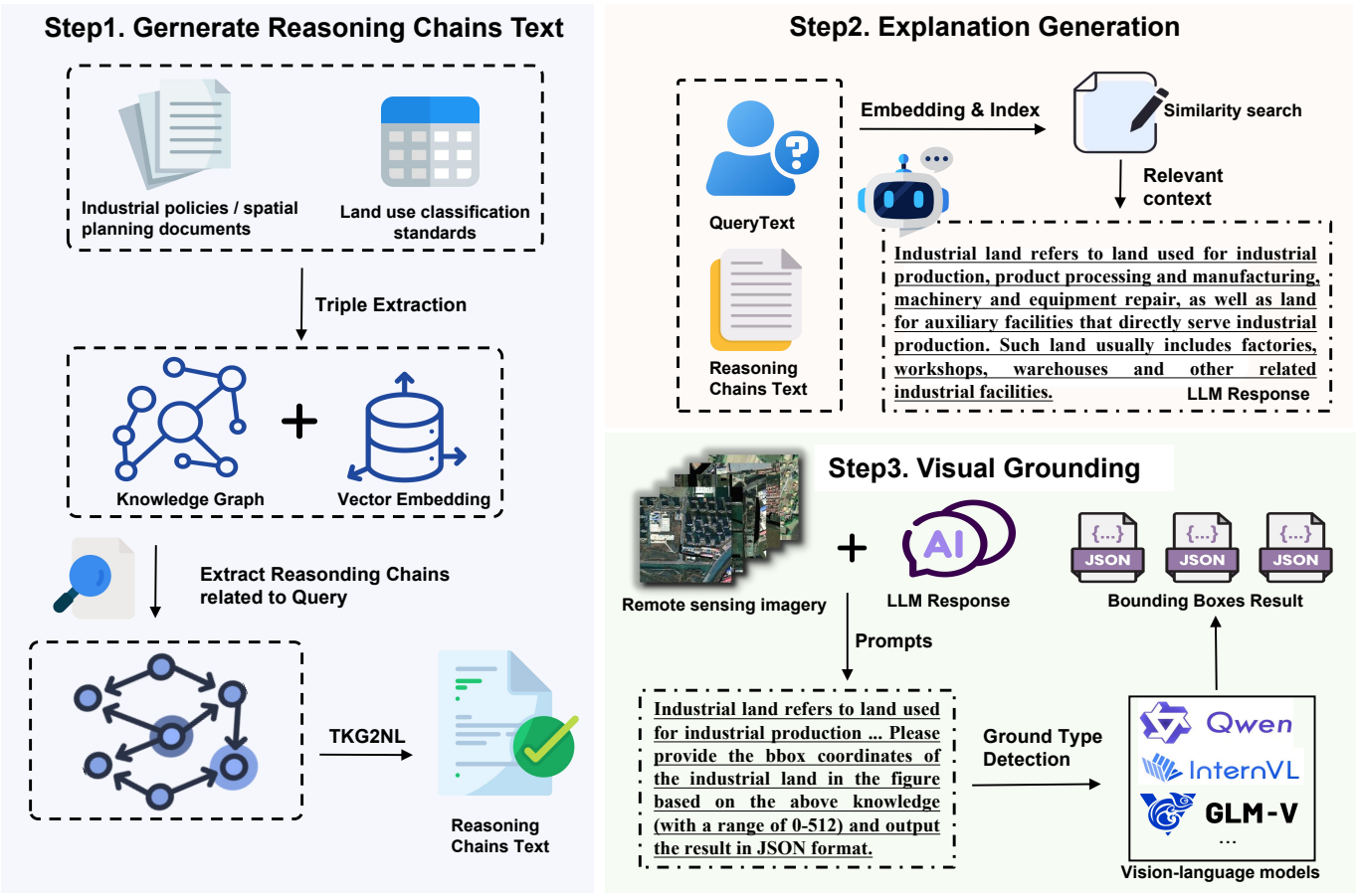


Fig. 5. **Detailed workflow of the RAGVLG framework.** Step 1 extracts triples from industrial policies and land-use standards to build the knowledge graph and vector embeddings, and converts query-relevant subgraphs into natural language reasoning chain text. Step 2 embeds query and reasoning texts, executes similarity search to retrieve domain context, and generates LLM explanatory responses. Step 3 feeds remote sensing images, LLM outputs and prompts into vision-language models for industrial land visual grounding, outputting bounding boxes in JSON format.

tations of visual features, which serve as queries to the Graph RAG retriever. The retrieved subgraph is then synthesized into domain-aware semantic feedback and cross-referenced with the VLM's visual embeddings, resulting in knowledge-informed, context-aware industrial land grounding. Specifically, we build the Graph RAG knowledge base fully relying on the Third National Land Survey Land Use Classification. We construct a standardized geospatial knowledge graph with three core graph components. Core entities cover industrial, mining and warehousing land and other classes, while auxiliary entities include spatially correlated residential, green and transportation land. Each entity stores official attributes including standardized land definitions, functional orientations, quantitative spatial indicators and planning control rules. Three relation types are defined for land spatial governance, covering hierarchical subordinate links, spatial topological constraints and functional causal interactions between industrial activities and supporting facilities. Benefiting from an adaptive expansion module, the framework automatically extracts structured knowledge from urban planning and ecological documents under a unified schema to enrich the domain knowledge base continuously. All graph triples are embedded and stored together with raw structured data in a vector database. When processing task queries targeting industrial land, the system first matches target core entities via vector search and implements multi-hop graph traversal along predefined relational links [43]. Redundant graph nodes are filtered to condense structured triples into concise, evidence-based domain explanations. Such refined land knowledge serves as fixed prior information embedded into reusable visual grounding prompt templates. These knowledge-augmented prompts are finally fed into the VLM alongside remote sensing imagery to complete industrial land visual grounding, where introduced domain rules mitigate spectral aliasing confusion and boost localization precision.

We evaluate framework performance by comparing:

- Baseline VLM performance (without RAG augmentation): We selected eight representative models for evaluation, including: 1) generalized VLMS: Qwen2.5-VL-72B-Instruct [44], Qwen2.5-VL-7B-Instruct [44], QVQ-72B-Preview [45], GLM-4.5V [46], and InternVL3-38B [47]; 2) remote sensing VLMs: GeoChat [48], LHRS-Bot [49] and TEOChat [50].

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- RAGVLG with different VLM backbones: The framework integrates the aforementioned VLMs as backbones and augments them with our Graph RAG module. For the core tasks of the Graph RAG module (graph extraction and question answering), we adopt models from Zhipu AI to ensure accuracy and efficiency. To standardize the Graph RAG module's performance, we configure its key parameters as follows: we use `iic/nlp_gte_sentence-embedding_chinese-base` as the embedding model; we set the minimum similarity score for k-nearest neighbor retrieval to 0.94 to filter high-relevance results; during graph construction, we combine 6 text chunks to enrich node information; and for graph updates, we process 20 chunks per iteration to balance update efficiency and data coverage.

To quantitatively evaluate the industrial land visual grounding performance of RAGVLG, we adopt F1-score as the core metric, which computes the harmonic mean of precision and recall. For box matching, we calculate Intersection over Union (IoU) between predicted and ground-truth bounding boxes with a threshold $\tau = 0.3$, where pairs are only valid for counting if their IoU meets this threshold and category labels match. We further apply a greedy maximum-IoU one-to-one matching scheme to avoid repeated counting [51]: all IoU pairs are sorted descendingly, and unoccupied valid pairs are marked as true positives while matched ground-truth boxes are excluded from subsequent matching. Unmatched ground-truth parcels are treated as false negatives, and redundant or unmatched predicted boxes are classified as false positives to derive rigorous precision and recall values for final F1 calculation. Subsequently, the model that performs the best in the RSIL dataset is selected for the applications of evaluating the industrial land of Shenzhen and Hong Kong.

C. Geographic-spatial assessment of potential environmental risks

Industrial land presents significant heat-related environmental risks due to its predominance of impervious surfaces, intensive energy consumption, and limited vegetation cover. These factors intensify the UHI effect, leading to elevated surface and ambient temperatures, disrupting local ecosystems, impairing human health, and diminishing agricultural productivity in adjacent areas [52].

Geographic-spatial analysis of such risks enables the identification of industrial parcels with higher environmental vulnerability, thereby providing a basis for targeted retrofitting interventions (e.g., increasing greenery, introducing shading, or improving surface materials). In this study, we focus on two structural parameters, thermal environment and vegetation coverage, to derive spatially explicit indicators: 1) local thermal anomalies, and 2) local greening gaps. These indicators were combined with confidence and quality-control thresholds to classify industrial land into risk tiers, prioritizing areas with the most prominent ecological issues. The ultimate aim is to support the optimization of industrial land use patterns and promote sustainable urban-industrial transformation

1) Thermal environment: Rapid urbanization alters both local climate and ecological processes, with the UHI phenomenon- higher urban air temperatures compared to rural surroundings being one of the most notable impacts [53], [54], [55], [56], [57]. Industrial land often functions as a major heat source within cities due to concentrated industrial activities, transportation networks, and extensive impervious surfaces. Waste heat emissions associated with these processes contribute significantly to elevated surface temperatures and the clustering of thermal hotspots [58], [59].

To characterize the thermal environment of Shenzhen and Hong Kong, we retrieved LST from Landsat 8 OLI/TIRS imagery [60] acquired in summer 2023 (June–August). For each industrial land bounding box, we computed the mean LST and defined a local thermal anomaly index ($\mathrm{LST}_{\mathrm{excess_local}}$) as:

$$\mathrm{LST}_{\mathrm{excess_local}} = \mathrm{LST}_{\mathrm{indus}} - \mathrm{LST}_{\mathrm{ring_mean}}. \quad (1)$$

where $\mathrm{LST}_{\mathrm{indus}}$ represents the average LST within the bounding box and $\mathrm{LST}_{\mathrm{ring_mean}}$ symbolize the average LST of the surrounding 500 m buffer zone (excluding the box interior). Positive values indicate parcels that are hotter than their immediate surroundings, thus representing higher heat risk.

2) Vegetation coverage: Urban vegetation plays a critical role in regulating local temperatures through shading and evapotranspiration [61]. Numerous studies have demonstrated that increased vegetation coverage significantly reduces urban temperatures, making it a key strategy for mitigating UHI effects [62], [63], [64].

To assess vegetation deficits within industrial land, we calculated NDVI from Landsat 8 OLI/TIRS summer composite imagery (June–August 2023). The NDVI was aggregated per bounding box and compared against the surrounding 500 m buffer zone. The local greening gap index ($\mathrm{NDVI}_{\mathrm{gap_local}}$) is defined as:

$$\mathrm{NDVI}_{\mathrm{gap_local}} = \max(0, \mathrm{NDVI}_{\mathrm{ring_mean}} - \mathrm{NDVI}_{\mathrm{mean_box}}). \quad (2)$$

where $\mathrm{NDVI}_{\mathrm{mean_box}}$ is the mean vegetation index within the bounding box and $\mathrm{NDVI}_{\mathrm{ring_mean}}$ is the mean NDVI in the surrounding 500 m buffer (excluding the box). Positive values indicate parcels with less vegetation than their local context, highlighting areas where ecological restoration and greening interventions could be prioritized.

3) Priority assessment: For robust risk delineation, we construct two indicators: confidence and priority score. Given the potential for imperfect recognition of industrial land bounding boxes, an initial screening step was introduced to identify hardened surfaces as industrial candidate pixels by applying an NDVI threshold σ, thereby generating an industrial candidate mask (i.e., NDVI value is smaller than σ). In particular, we define the value of $\mathrm{LST}_{\mathrm{indus}}$ by setting a threshold for the proportion of industrial candidate pixels within each bounding box ($\mathrm{Indus}_{\mathrm{frac}}$). When $\mathrm{Indus}_{\mathrm{frac}} \geq 0.2$, $\mathrm{LST}_{\mathrm{indus}}$ is assigned the average LST within the industrial candidate mask; otherwise, it is replaced with the average LST of the entire bounding box.

For confidence calculation, the proportion of industrial candidate pixels within each bounding box ($\mathrm{Indus}_{\mathrm{frac}}$) was used as a proxy for the likelihood that the detected bounding

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box corresponds to actual industrial land. This proportion was converted into a confidence weight according to:

$$w_{\text{conf}} = \min\left(1, \frac{\text{Indus}_{\text{frac}}}{0.6}\right). \tag{3}$$

When $\text{Indus}_{\text{frac}}0.6$, the confidence weight (w_{conf} is scaled proportionally to the ratio of actual fraction to 0.6. For bounding boxes where $\text{Indus}_{\text{frac}} \geq 0.6$, the confidence weight is set to 1.

Based on the localization confidence weight ω_{conf} derived from Equation 3, we calculate parcel-wise environmental priority scores via weighted aggregation of two percentile-normalized ecological pressure metrics, $\text{rank}(\text{LST}_{\text{excess_local}})$ and $\text{rank}(\text{NDVI}_{\text{gap_local}})$. Both ranked metrics fall within the $[0, 1]$ interval, and percentile ranking eliminates dimensional discrepancies between Celsius-based LST measurements and dimensionless NDVI to ensure urban-scale cross-parcel comparability. Weighting coefficients ω_{LST} and ω_{NDVI}, subject to the constraint $\omega_{\text{LST}} + \omega_{\text{NDVI}} = 1$, quantify the respective ecological stress from thermal anomalies and vegetation deficits; we adopt $\omega_{\text{LST}} = 0.7$ and $\omega_{\text{NDVI}} = 0.3$ to attach greater priority to industrial heat-island impacts. The composite scoring formulation is expressed as:

$$\begin{aligned}\text{Priority} = w_{\text{conf}} \times \Big[&w_{\text{LST}} \cdot \text{rank}(\text{LST}_{\text{excess_local}}) \\ &+ w_{\text{NDVI}} \cdot \text{rank}(\text{NDVI}_{\text{gap_local}})\Big].\end{aligned} \tag{4}$$

The resulting environmental risk priority scores were then classified utilizing the K-means clustering algorithm [65] into five gradient categories: extremely high risk, high risk, medium risk, low risk, and extremely low risk. This multi-tier classification captures subtle variations in thermal stress and vegetation deficit across industrial parcels, providing a refined decision-making basis for developing targeted and differentiated ecological management strategies.

4) Spatial clustering: Spatial clustering of industrial environmental risk was assessed using Local Moran's I (Local Indicators of Spatial Association, LISA)[66]. The analysis was conducted on 1 km × 1 km regular grid cells for Shenzhen and Hong Kong. Spatial adjacency was defined via the Queen contiguity rule (sharing edges or vertices), and row-standardized spatial weight matrix was built for spatial calculation. Significance was verified at the threshold of $p < 0.05$. The formula of Local Moran's I_i for grid unit i is specified as:

$$I_i = \frac{x_i - \bar{x}}{S^2} \sum_{j=1}^{n} w_{ij}(x_j - \bar{x}). \tag{5}$$

where x_i denotes area-weighted industrial risk value of grid i, $\bar{x}$ is the overall average risk across all grids, w_{ij} refers to spatial weight between grid i and grid j, and S^2 stands for sample variance:

$$S^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2. \tag{6}$$

Positive I_i corresponds to significant High–High (HH) or Low–Low (LL) spatial clustering, while negative I_i represents spatial outliers including High–Low (HL) and Low–High (LH) patterns; non-significant units are unclassified. This classification quantitatively delineates the spatial distribution of industrial risk and further unpacks the driving effects of urban planning and geographic conditions.

IV. Results

A. Results of identifying Industrial land

1) Demonstration of the effectiveness of our approach on different VLMs: A principal contribution of the proposed framework is that it is model-agnostic, as RAGVLG can be seamlessly adapted to various mainstream VLM backbones without requiring architectural modifications. Quantitative (see Table I) evaluations demonstrate that, for each evaluated VLM backbone, RAGVLG yields superior results compared to corresponding baseline implementations without retrieval augmentation. This confirms that the framework's knowledge-informed visual grounding is not restricted to a particular model design but instead serves as a general enhancement strategy for industrial land recognition. Among the evaluated models, Qwen2.5-VL-72B-Instruct with RAGVLG attains the highest total F1 score, representing a 4.43% improvement over its baseline. This gain is driven primarily by a 5.38% increase in recall, indicating enhanced capability to detect industrial land parcels while maintaining comparable precision.

Figure 6 shows the visual results of different VLMs with RAGVLG. The performance of Qwen2.5-VL-72B-Instruct with RAGVLG is particularly pronounced in complex urban mixed-use scenarios, where visual ambiguity and overlapping features often degrade detection accuracy. Given its balanced precision–recall profile and robustness, Qwen2.5-VL-72B-Instruct with RAGVLG was selected as the backbone for subsequent large-scale recognition and geospatial risk assessment experiments.

2) Ablation study: Graph RAG vs. Standard RAG: To evaluate the specific contribution of Graph RAG within the RAGVLG framework, we conducted an ablation study using Qwen2.5-VL-72B-Instruct as the baseline model. Two configurations were tested: (1) standard RAG retrieval and (2) Graph RAG retrieval. In both cases, the Third National Land Survey classification table served as the domain knowledge source. As shown in Table II, Graph RAG improves the F1 score by 3.08% compared to standard RAG under the same IoU setting (0.3). This improvement is largely attributable to a 3.59% gain in recall, with a modest precision increase of 0.20%. The performance benefit results from Graph RAG's structured knowledge representation, namely the entity–attribute–relationship graph, which preserves semantic relationships among industrial land attributes and supports multi-hop reasoning. In contrast, standard RAG relies primarily on keyword matching, which often fails to capture relational logic within domain-specific data, leading to incomplete or mismatched retrievals.

Figure 7 compares three methods: RAGVLG (ours), Standard RAG, and no-RAG (baseline), in identifying industrial land bounding boxes from remote sensing imagery. RAGVLG uses graph-structured knowledge to generate more informative queries that capture typical industrial features (e.g., large buildings, warehouses), leading to more complete detection.

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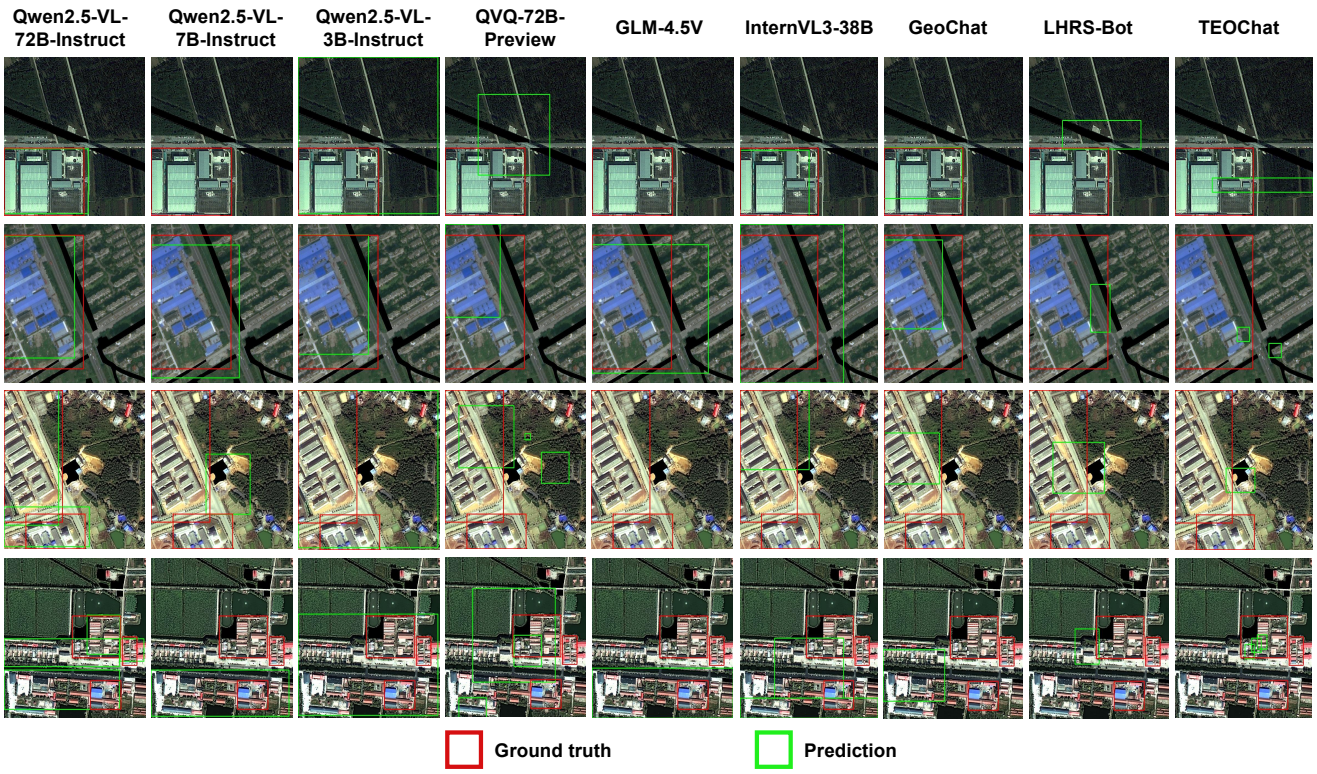


Fig. 6. Visual grounding results of different VLMs with RAGVLG on RSIL dataset

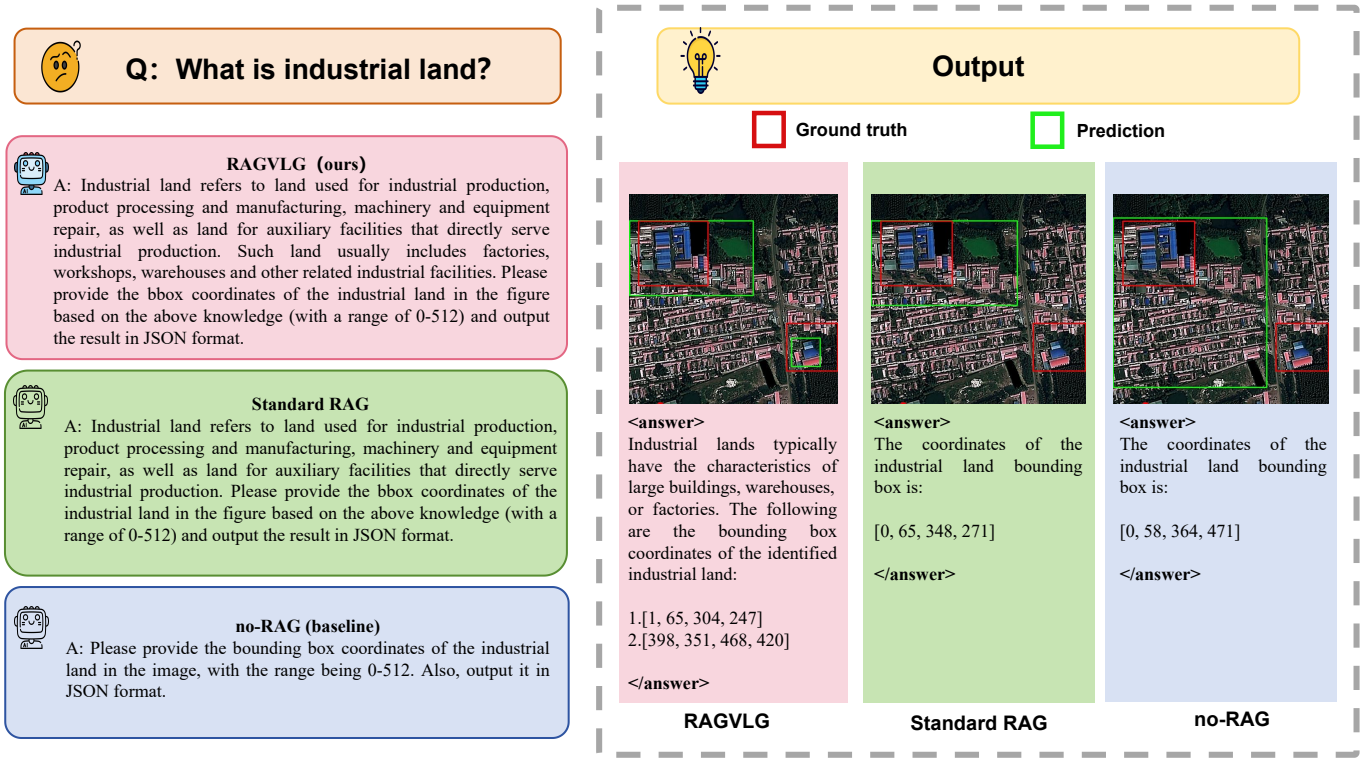


Fig. 7. **Prompt and output examples of different RAG methods**. The comparison reflects knowledge guidance's influence on grounding accuracy. RAGVLG detects all scattered industrial plots with superior recall. Standard RAG misses small targets and outputs oversized bounding boxes. The no-RAG baseline misidentifies residential areas due to absent domain knowledge constraints.

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT VLM BACKBONES WITH AND WITHOUT RAGVLG ON THE RSIL DATASET.

Vision language model	Precision		Recall		F1 score	
	Baseline	RAGVLG	Baseline	RAGVLG	Baseline	RAGVLG
Qwen2.5-VL-72B-Instruct	0.3091	0.2899	0.1208	0.1746	0.1737	0.2180
Qwen2.5-VL-7B-Instruct	0.2032	0.2050	0.0515	0.0521	0.0822	0.0831
QVQ-72B-Preview	—	0.1547	—	0.0405	—	0.0642
GLM-4.5V	0.1882	0.2270	0.0101	0.0200	0.0192	0.0367
InternVL3-38B	0.2171	0.1927	0.0683	0.0837	0.1040	0.1167
GeoChat	0.2270	0.2455	0.0910	0.0988	0.1299	0.1409
LHRS-Bot	0.0547	0.0457	0.0104	0.0148	0.0175	0.0223
TEOChat	0.0417	0.0353	0.0310	0.0423	0.0363	0.0385

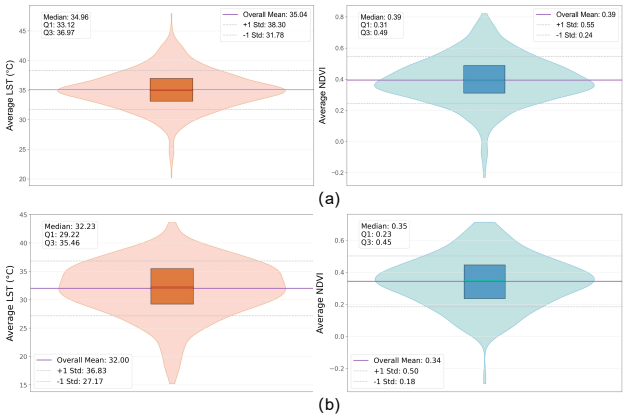


Fig. 8. Distributions of average LST (left) and average NDVI (right) values for industrial land parcels. (a) Shenzhen; (b) Hong Kong.

As shown, RAGVLG correctly identifies two distinct industrial zones (red for ground truth and green for prediction), while Standard RAG and no-RAG miss the secondary area (e.g., in the lower-right), demonstrating that integrating structured knowledge helps reduce missed detections.

B. Geographic-spatial analysis of industrial land

To further demonstrate the practical applicability of RAGVLG, we carried out comprehensive case studies in two megacities: Shenzhen and Hong Kong. Rapid industrial expansion in Shenzhen induces prominent urban heat and vegetation loss, while Hong Kong's concentrated port and reclamation-based industries cause similar thermal risks. Their divergent urban layouts enable cross-environment validation of the framework.

We applied the optimized Qwen2.5-VL-72B-Instruct + RAGVLG configuration to high resolution remote sensing imagery of Shenzhen and Hong Kong for city-scale industrial land identification. In Shenzhen, the model processed 1,000 PlanetScope image patches, each measuring 512×512 pixels, and detected 1,265 industrial land bounding boxes. Manual visual interpretation confirmed 1,100 correct industrial parcels, yielding a precision of 87%. In Hong Kong, 495 satellite image patches were processed, producing 551 detected industrial land bounding boxes, of which 470 were verified as correct, corresponding to a precision of 85%. For subsequent environmental risk assessment, all bounding box coordinates were transformed into the projected coordinate system of the LST and NDVI datasets to ensure spatial alignment. The mean LST and NDVI values were then extracted for each verified industrial parcel.

The statistical distribution of all industrial land parcels in Shenzhen was visualized using violin plots (Figure 8), providing an intuitive representation of parcel-level environmental conditions, offering spatial references for subsequent differential analyses. In Shenzhen, average LST values cluster around a median of 34.96°C, with an overall mean of 35.04°C and a range of approximately 20–45°C; average NDVI centers on a median of 0.39, matching the overall mean, with values spanning roughly -0.20 to 0.80. In Hong Kong, average LST is lower overall, with a median of 32.23°C and an overall mean of 32.00°C, while NDVI is slightly lower, with a median of 0.35 and an overall mean of 0.34. Both cities show similar distribution patterns, but Shenzhen's industrial parcels exhibit higher thermal conditions and marginally higher vegetation coverage compared to Hong Kong's.

Figure 9 presents the spatial distribution of industrial land-related environmental risk, with darker colors indicating higher risk levels. The map highlights distinct high-risk industrial agglomerations and priority intervention zones in both cities. In Shenzhen, the highest-risk clusters are mainly concentrated in Nanshan and Bao'an Districts. Nanshan, characterized by technology-intensive manufacturing and dense built-up development, exhibits elevated risk due to the combined effects of heat accumulation and limited green space. Bao'an, which contains extensive traditional manufacturing areas, similarly shows persistent thermal accumulation and insufficient vegetation coverage. In Hong Kong, high-risk industrial land is concentrated primarily in Kwai Tsing District, where port and logistics activities are densely distributed, and in the Outlying Islands District, particularly around Hong Kong International Airport on Lantau Island. In these areas, extensive impervious surfaces and limited vegetation contribute to persistent heat accumulation. Overall, the spatial pattern suggests that industrial environmental risk in both cities is closely associated with

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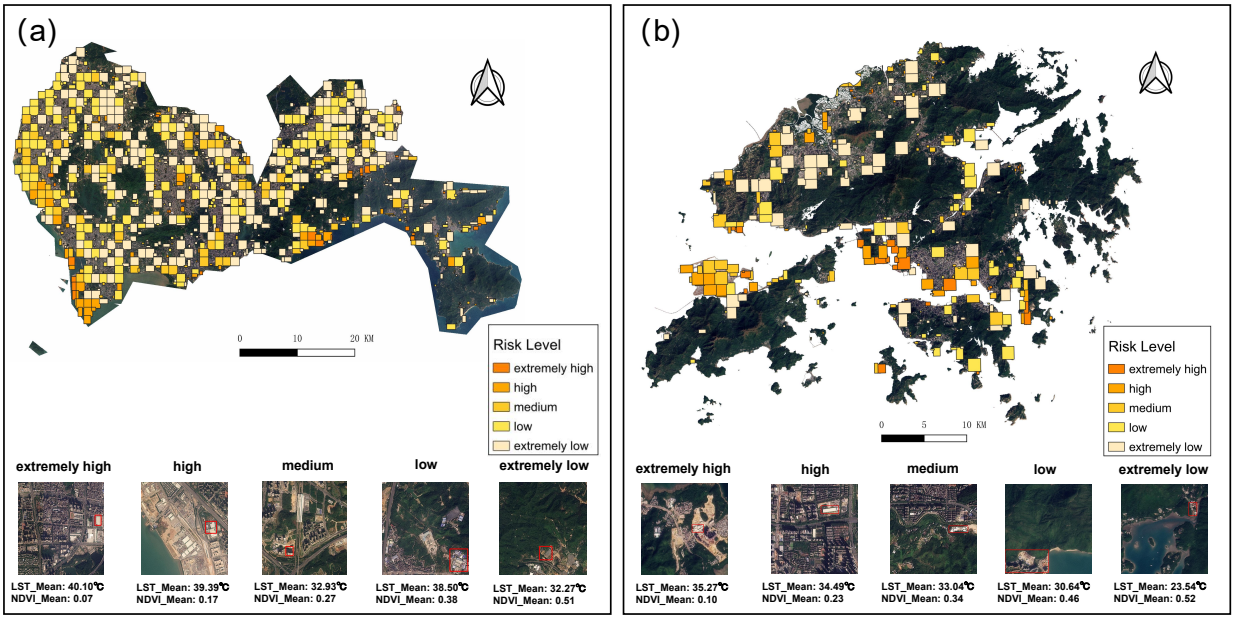


Fig. 9. **Risk distribution of industrial land in Shenzhen (a) and Hong Kong (b), with five risk levels shown alongside typical samples and their corresponding LST/NDVI values.** The typical samples illustrated below demonstrate the empirical coupling between risk levels and environmental metrics: extremely high/high risk zones are prominently coupled with severe thermal anomalies (high LST means) and bare or densely built concrete infrastructure (low NDVI means), whereas low/extremely low risk zones are characterized by well-mitigated microclimates and abundant vegetation coverage.

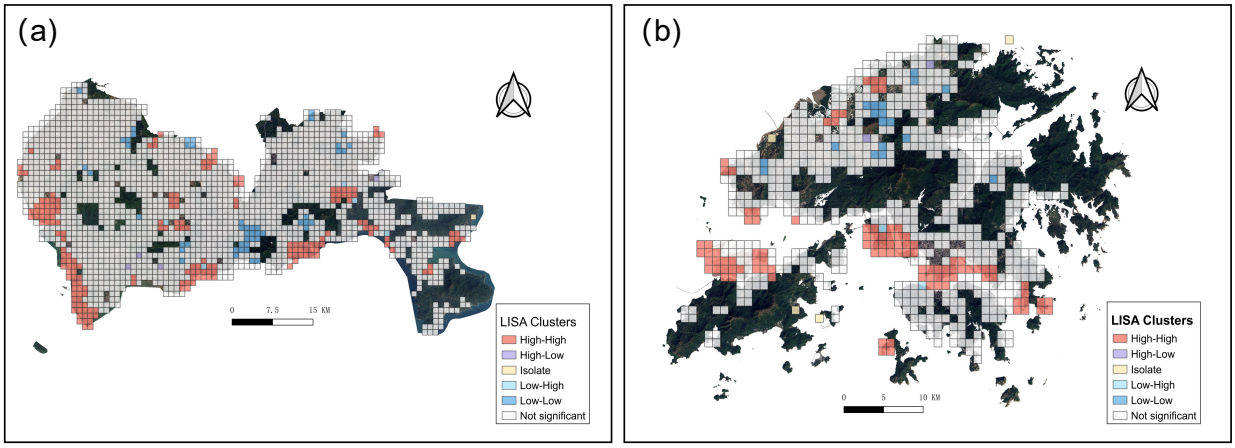


Fig. 10. **LISA cluster maps of industrial risk in Shenzhen (a) and Hong Kong (b)**. The maps clearly reveal the spatial risk patterns: prominent High-High clusters are directly observed along the coast of Shenzhen and the Kwai Tsing-Kowloon area in Hong Kong, highlighting critical zones requiring urgent environmental restoration, while Low-Low clusters pinpoint sustainable eco-friendly industrial areas.

TABLE II
PERFORMANCE COMPARISON OF RAG METHODS

	Baseline	Standard RAG	RAGVLG
Precision	0.3091	0.2879	0.2899
Recall	0.1208	0.1387	0.1746
F1 score	0.1737	0.1872	**0.2180**

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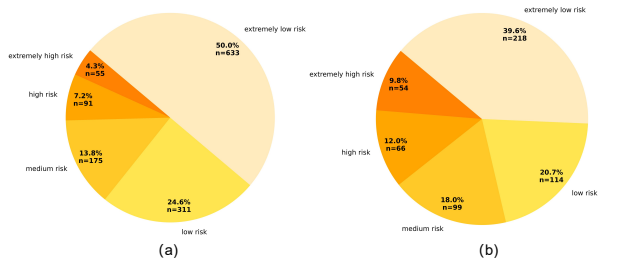


Fig. 11. Distribution of industrial land parcels by risk level. (a) Shenzhen; (b) Hong Kong.

thermal conditions and vegetation coverage, underscoring the need for targeted heat mitigation and greening interventions.

The LISA cluster maps in Figure 10 further reveal marked differences in the spatial agglomeration of industrial environmental risk between Shenzhen and Hong Kong. HH hotspots account for a larger proportion of industrial grids in Hong Kong (15.31%) than in Shenzhen (10.62%), although their spatial configurations differ substantially. In Shenzhen, HH clusters form a relatively continuous coastal risk corridor extending across Bao'an, Nanshan, Futian, Luohu, and Yantian. This pattern is closely related to the city's history of large-scale and spatially contiguous industrial park development during its early industrialization period. By contrast, Hong Kong's HH hotspots are more spatially fragmented, with three major clusters located in Kwai Tsing, Kowloon, and the Outlying Islands. Constrained by mountainous terrain and limited developable land, port-oriented and logistics-related industrial activities in Hong Kong are concentrated within fragmented reclaimed and lowland areas, resulting in a higher HH ratio despite a discontinuous spatial pattern. For LL coldspots, Shenzhen and Hong Kong show nearly identical overall proportions, accounting for 2.75% and 2.76% of industrial grids, respectively. However, their spatial distributions are notably different. In Shenzhen, LL cells are mainly concentrated in inland core urban areas, forming a relatively consolidated low-risk region. This pattern may reflect the effects of long-term urban renewal, industrial relocation, and improvements in urban greening. In contrast, Hong Kong's LL coldspots are sparsely distributed across mountainous countryside and ecological conservation areas, reflecting the influence of rugged terrain, strict land-use controls, and the limited availability of developable land. These contrasting cluster configurations indicate that industrial risk agglomeration is shaped not only by industrial land intensity but also by broader differences in urban development trajectories, industrial land organization, natural terrain, and land management systems.

To further quantify the prevalence of different risk categories, we calculated the number and proportion of industrial land parcels at each risk level, as shown in Figure 11. In Shenzhen, the number of parcels increases as the risk level decreases. Extremely high-risk parcels account for only 4.3% of all industrial parcels, corresponding to 55 parcels, whereas extremely low-risk parcels account for 50.0%, corresponding to 633 parcels. This distribution suggests that most identified industrial land parcels in Shenzhen are associated with relatively low environmental risk. By contrast, Hong Kong exhibits a higher concentration of parcels in high-risk categories. Extremely high-risk parcels account for 9.8%, or 54 parcels, and high-risk parcels account for 12.0%, or 66 parcels, while extremely low-risk parcels represent a lower share of 39.6%, or 218 parcels. Taken together, these results indicate that Hong Kong's industrial land has a notably higher proportion of high-risk parcels than Shenzhen, reflecting greater environmental risk exposure within its industrial zones.

V. Discussion

A. Methodological significance

This study develops a retrieval-augmented vision–language framework for industrial land visual grounding in remote sensing imagery. The RAGVLG framework is evaluated on the RSIL dataset and further applied to city-scale industrial land identification in Shenzhen and Hong Kong. By integrating industrial land maps with thermal environment and vegetation coverage indicators, the study generates spatially explicit risk maps and reveals the spatial distribution of industrial environmental vulnerability.

The proposed workflow provides practical decision support for industrial land management. Integrating thermal anomaly and vegetation deficit features, it hierarchically identifies industrial parcels with varying heat accumulation and greening deficiency levels, enabling differentiated ecological renovation and regulatory strategies across graded risk zones. Extremely high-risk parcels with severe heat island effects and insufficient green space are prioritized for emergency ecological intervention, including building vertical and rooftop greening [28], [67], illegal structure clearance, and pocket park construction [68]. High-risk industrial zones adjacent to residential areas require targeted industrial upgrading and buffer optimization, which guides the relocation of high-energy-consumption and high-pollution enterprises and mandates 50-100-meter green ecological buffer zones to mitigate human-environment conflicts [69], [70], [71]. Medium-risk parcels are designated for routine environmental monitoring, with strict constraints on greening ratios and heat emissions for subsequent factory expansion approvals to prevent risk degradation. For low and extremely low-risk industrial zones with superior environmental conditions, their optimal building density and greening indicators can be summarized as standardized planning benchmarks for urban industrial layout; extremely low-risk regions are further incorporated into ecological protection red lines to restrict high-intensity development [72]. Additionally, the derived thermal and vegetation indicators establish quantitative environmental baselines, supporting the formulation of parcel-specific renovation targets and standardized post-intervention monitoring. Under extreme heat weather, this hierarchical risk classification also facilitates precise resource allocation and targeted heat-mitigation adaptation measures.

Overall, the workflow links remote sensing-based environmental assessment with transformation-oriented decision-making. It offers a replicable approach for ecological management, industrial layout optimization, and sustainable urban-industrial development.

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B. Research innovation

This study contributes to industrial land visual grounding and environmental risk assessment by improving domain awareness, transferability, and application relevance in remote sensing interpretation.

The proposed RAGVLG framework advances conventional VLM-based approaches by incorporating retrieval-augmented domain knowledge into the grounding process. Existing VLMs typically rely on end-to-end visual inference and may misclassify visually similar or semantically ambiguous land-use features in complex urban-industrial environments [48], [49], [50]. RAGVLG addresses this limitation by retrieving industrial land semantic attributes, interpretation rules, contextual exemplars, and spatially relevant descriptions to guide model reasoning. This knowledge-informed process helps reduce domain ambiguity and improves industrial parcel localization.

The framework is also designed to be modular and backbone-agnostic. Many remote sensing visual grounding models are tightly coupled with specific architectures or task-specific training pipelines [73], [31], [74], making adaptation to different VLMs costly and less scalable. In RAGVLG, the retrieval module operates independently of the VLM backbone and can be integrated with different mainstream models without substantial architectural modification. This design supports consistent performance improvement and facilitates deployment across broader geospatial applications.

Beyond methodological improvement, this study extends visual grounding from controlled datasets to city-scale environmental analysis. Existing methods are often validated on small-scale or relatively homogeneous datasets, which may not fully capture the complexity of real-world urban environments [75], [76], [77], [78]. By applying RAGVLG to Shenzhen and Hong Kong, this study demonstrates its capacity for large-scale industrial land mapping under heterogeneous urban conditions. The integration of thermal and vegetation indicators further converts grounding results into quantitative spatial evidence for industrial planning and environmental risk management.

C. Comparative implications for Shenzhen and Hong Kong

The differences in industrial risk patterns between Shenzhen and Hong Kong reflect their distinct urban development trajectories, industrial land organization, terrain constraints, and planning regimes. Shenzhen's legacy of large-scale and contiguous industrialization has produced an extensive coastal risk corridor, while urban renewal, industrial upgrading, and relocation have contributed to lower-risk areas in parts of the urban interior [79], [80], [81]. This pattern suggests that Shenzhen may benefit from corridor-scale mitigation strategies, including coastal industrial belt greening, thermal environment regulation, and ecological buffering.

Hong Kong presents a more compact and fragmented risk pattern. Persistent land scarcity, mountainous terrain, limited developable land, and strict planning controls have concentrated high-risk industrial and logistics functions in a small number of intensively used areas, especially port-related zones and older industrial districts [82], [83]. These compact hotspots may generate higher localized exposure where industrial activities are close to dense residential or commercial areas. Risk mitigation in Hong Kong therefore requires more localized interventions, such as greening compact industrial districts, improving ventilation corridors, using reflective materials, and redeveloping obsolete industrial buildings.

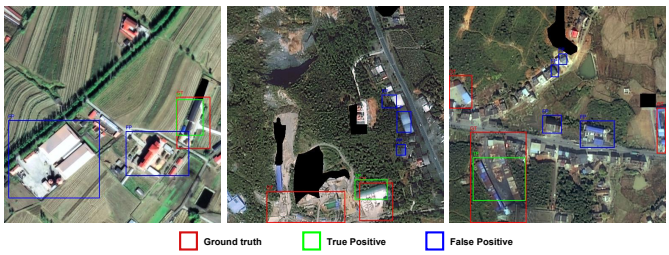


Fig. 12. Qualitative visualization of industrial land grounding outputs.

This comparison shows that industrial environmental risk is not determined by industrial land presence alone. It is jointly shaped by industrial morphology, urban density, green space configuration, terrain constraints, development history, and institutional planning. Incorporating both environmental indicators and urban spatial context is therefore essential for robust industrial land risk assessment.

D. Limitations and future work

Despite the reliable industrial land grounding performance of the RAGVLG framework, false positive detections remain prevalent in complex urban scenarios, as visually exemplified by the representative detection cases in Figure 12. Such misdetections are primarily attributed to visual feature aliasing. Urban facilities including warehousing centers, large commercial buildings, and low-density residential areas share analogous textures and parcel scales with industrial plots, triggering visual confusion in visual-language model inference. Future work will adopt visual contrastive learning to alleviate such feature confusion, reduce false positive rates, and further improve the overall model robustness.

The RSIL validation dataset focuses exclusively on industrial land, which supports targeted optimization but limits the evaluation of cross-category generalization. Future studies could include residential, commercial, agricultural, logistics, and public service land to assess the transferability of RAGVLG across land-use types.

The retrieval module and knowledge base are designed mainly for industrial land interpretation. Extending the framework to other land-use categories or geographic contexts will require more flexible knowledge organization, customized retrieval logic, and region-specific interpretation rules. Moreover, the current knowledge base is oriented toward the Chinese urban-industrial context, which may limit cross-border applicability. Incorporating global land-use ontologies, multilingual planning documents, international zoning knowledge, and cross-regional remote sensing exemplars would improve broader transferability.

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The environmental risk assessment also focuses mainly on thermal environment and vegetation coverage. Although these indicators capture important dimensions of industrial environmental vulnerability, future work could integrate air pollution, soil contamination, hazardous material storage, flood exposure, energy consumption, proximity to residential areas, and socioeconomic vulnerability to support a more comprehensive risk assessment.

VI. Conclusion

Industrial activity, while foundational to urban economic vitality, exacerbates the UHI effect through concentrated anthropogenic heat emissions and impervious surfaces that trap thermal energy. These thermal imbalances undermine ecological stability, heighten public health risks, and increase energy demand, which pose barriers to achieving the United Nations SDGs. Consequently, precise identification of industrial land and systematic thermal assessment are crucial to targeted heat mitigation and to guiding sustainable urban-industrial planning.

This study addresses these challenges by constructing a dedicated RSIL dataset of 1,389 high-resolution images with detailed ground-truth annotations. It introduces RAGVLG, a retrieval-augmented vision-language framework that improves industrial land grounding accuracy via context-aware, knowledge-informed inference. Applying RAGVLG in Shenzhen and Hong Kong enabled large-scale mapping of industrial parcels and integration of thermal and vegetation indices to assess environmental risk spatially. The resulting risk mapping not only identifies concentrated high-risk zones but also supports a structured decision-support system for ecological retrofits, dynamic monitoring, and climate adaptation in industrial districts. The methodological architecture of RAGVLG is replicable, scalable, and inherently model-agnostic, rendering it suitable for broader applications in urban-industrial ecological governance. By combining cutting-edge vision-language modeling with practical environmental management, this research advances remote sensing analytics and operationalizes sustainable industrial land development. It ultimately aligns industrial transformation with SDG-driven urban resilience goals.

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